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Stratigraphic Evolution of the Middle to Upper Jurassic Gotnia Platform Margin, Saudi Arabia

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Abstract

Platform-to-basin correlations require an integrated approach due to the rapid lateral changes in facies, stratal geometries, environments, and biota. The Gotnia Basin is characterized by the presence of world-class source-rocks, while reservoir-seal pairs are found across the depositional profile in the platform, slope, and basin (Yousif and Nouman, 1997; Sharland et al. 2001; Ziegler, 2001). Developing a genetic understanding of platform-margin evolution is crucial for predicting elements of the petroleum system including reservoir-seal distribution.

In this study, a stratigraphic cross-section is constructed from platform to basin for the Middle to Upper Jurassic Gotnia Basin margin in Saudi Arabia. Sequence boundaries and maximum flooding surfaces are defined from well logs, core, and observed seismic geometries. A robust biostratigraphic framework exists on the platform, whereas reliable age-control in the basin is limited, primarily due to the highly restricted, hypersaline nature of the basin sediments. By combining these multiple data-types, a stratigraphic cross-section is constructed, capturing depositional facies, stratal geometries, and chronostratigraphic relationships.

From the Bajocian to Callovian, the marine shelf undergoes a transition from a low-angle mud-dominated ramp to a high-rise platform with distinct margin and slope geometries, characterized by an estimated vertical relief of over 200 metres. Multiple cycles of basin restriction and desiccation occur during the Kimmeridgian, most likely initiated by small eustatic sea-level drops, but with the majority of the sea-level fall attributed to evaporitic drawdown due to restricted seaway connections, resulting in the deposition of thick intervals of halite. Sylvite stringers within the halite provide evidence for complete evaporation. These evaporite packages correlate with exposed sequence boundaries on the platform top. During the Tithonian, the deposition of peritidal and evaporite-dominated sabkha on the platform and correlative carbonate muds and sub-aqueous evaporites in the basin infill the high-rise platform topography, modifying the depositional profile from platform back to ramp.

The Middle Jurassic to Lower Cretaceous thickness map exhibits near-isopachous characteristics across the platform margin, suggesting that the high-rise margin architecture that occurs in the Callovian to Oxfordian is the result of spontaneous, self-organizing carbonate growth rather than a response to fault-

related subsidence. Core in the basin reveals that autochthonous biostromal stromatolite and sabkha deposits develop between halite accumulations in a lowstand position, potentially forming effective reservoirs.

Introduction

Carbonate platform margin-to-basin transitions are fundamentally difficult to correlate across. Depositional environments change dramatically from platform-margin to the basin so lithostratigraphy names and definitions are often very different and provide poor guidance in how to identify time-correlative units. Sea-level highstands are well preserved on the platform as thick shallowing-up marine deposits, but they are often poorly represented in the basin. Conversely, sea-level lowstands are typically represented by non-deposition and exposure on the platform top, while thick sedimentary deposits may accumulate in the basin. [Gravestock et al. \(2021\)](#) shows a regional well-based correlation that covers the Gotnia Margin, while [Wharton \(2017\)](#) used seismic correlation to illustrate the geometries of the Gotnia Margin in this study area.

A time-correlative understanding of sedimentary evolution is important in the exploration industry, because a process-based genetic understanding of the stratigraphy allows us to make predictions of the distribution of reservoir, seals and source-rock that go beyond the existing subsurface data.

Biostratigraphy is one tool that allows us to correlate from platform to basin, but the fauna and flora on which we base biozonations can change from the shallow platform (dominantly benthic organisms) to the basinal environment (dominantly pelagic organisms). [Hughes and Naji \(2009\)](#) published biostratigraphic data for the restricted to marine flooding transition of the Hith Formation and the Manifa Member, while [Kadar et al. \(2015\)](#) published biostratigraphic data from the basinal section in Kuwait, and proposed a correlation with the platform sediments.

Observations from seismic data are often very useful to understand correlations across the margin, but these are limited by resolution, and seismic reflectors frequently converge in by-pass surfaces on a slope-margin, making time-correlations non-unique. For example, an exposure surface (a sequence boundary) on the platform top (a single seismic reflector) may thicken into multiple thick sedimentary packages in the basin. Conversely, a single reflector at the maximum flooding surface in the basin may be time-correlative to multiple thick highstand packages on the platform top.

Construction of a stratigraphic well cross-section that incorporates biostratigraphic control, seismic observations, and a coherent sequence stratigraphic framework is the best workflow for building a reliable platform-to-basin correlation scheme. Applying sequence stratigraphy means identifying sequence boundaries and maximum flooding surfaces, and recognizing sedimentary stacking patterns, and changes in depositional style. It also means understanding the hierarchy of the sequences observed, and its relationship to time (from biostratigraphic data).

In this study the platform stratigraphy and ages are well-constrained ([Powers, 1968](#), Aramco Stratigraphic Column, 2025). The basinal stratigraphy is also well defined, except around the evaporitic sections of the Gotnia and Hith Formations. During these times hypersaline environments develop, in which few organisms can survive and there is a stratigraphic window with very poor biostratigraphic control. [Kadar et. al \(2015\)](#) identify the youngest pre-evaporite as Kimmeridgian (biozone NJT14), and the oldest post-evaporite age in the Makhul as Tithonian (biozone NJT15-17, marked by the arrival of *C. mexicana minor*), placing the Gotnia Formation as Kimmeridgian to Tithonian in age. This leaves multiple options for correlating Gotnia salt cycles to the Jubaila-Arab-Hith platform sequences.

Aims

The aim of this study is to construct an internally consistent and robust correlation framework for the Middle to Upper Jurassic Gotnia Margin platform to basin transition in the study area. The model seeks to respect the biostratigraphic data, the geometries observed from seismic, and the facies distribution and evolution as

observed from logs and core. Figure 1 shows the regional location of the study area on the southern margin of the Gotnia Basin. Figure 2 shows the platform type-stratigraphy, including the ages of the formations.

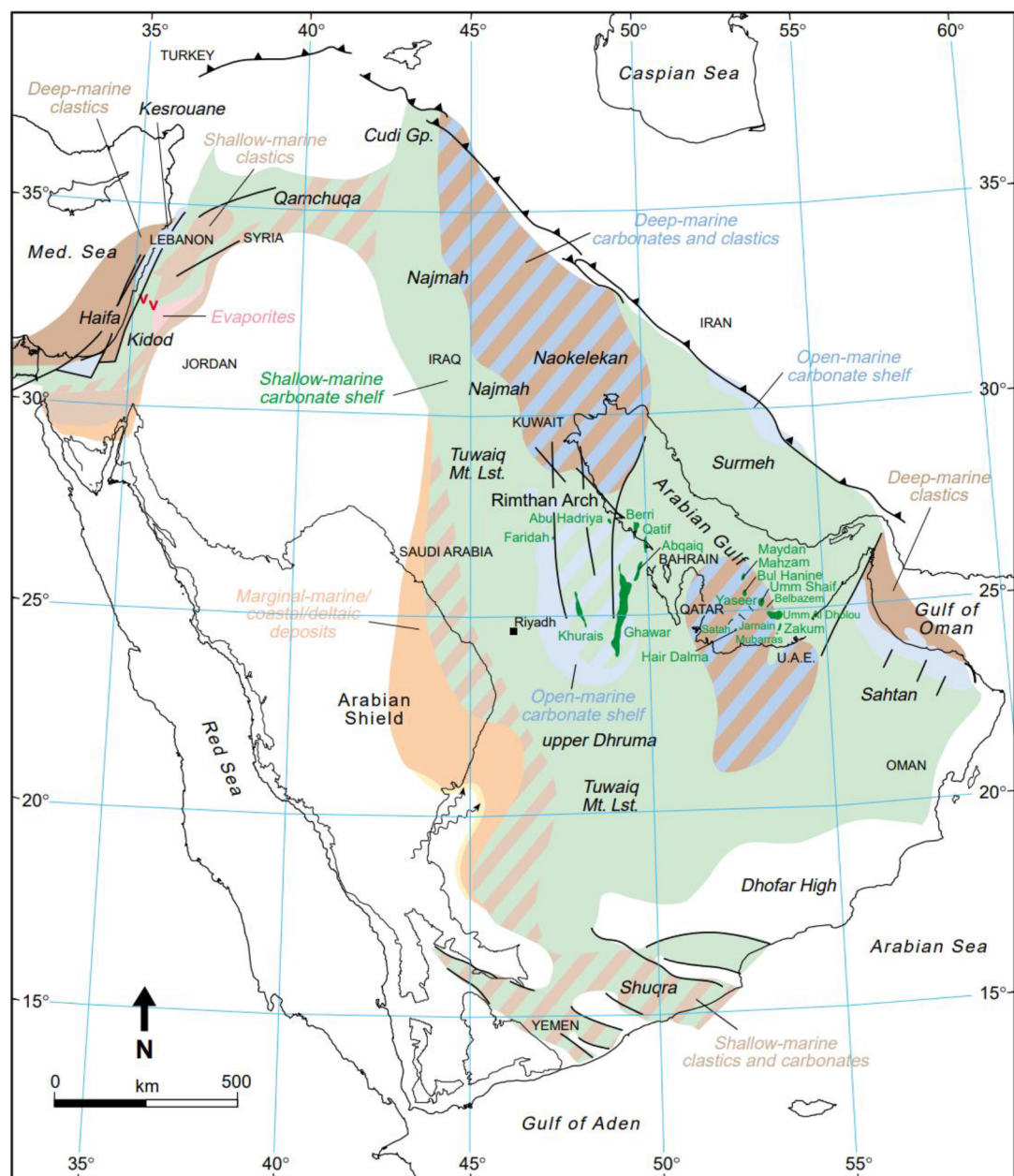


Figure 1—Map shows the depositional environments during Callovian to Oxfordian times (depositional environments map modified from Ziegler 2001).

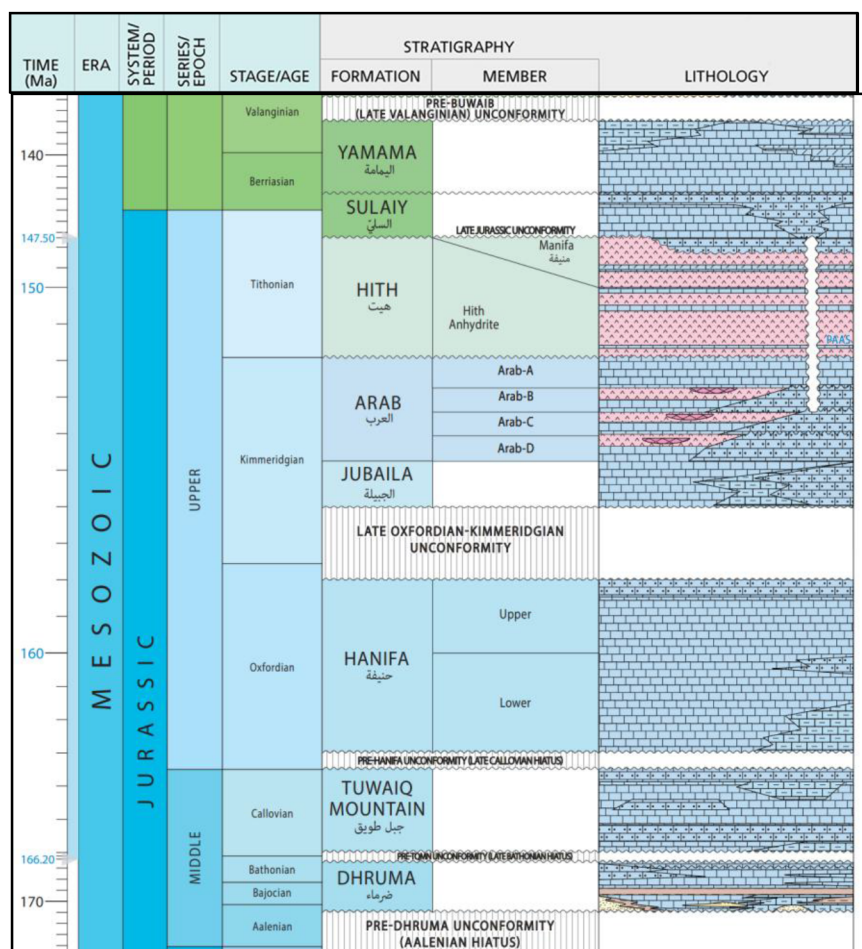


Figure 2—Stratigraphic column showing the study interval (modified from the Aramco Stratigraphic Column 2025).

Given the inherent uncertainties in the biostratigraphic ages, and the seismic resolution, the correlations proposed here are not unique, and other interpretations exist. However, the model proposed here is the authors' preferred model until additional data (biostratigraphy in the basin) is available.

Method

First a dip-section of closely spaced wells is chosen. Figure 3 shows a map indicating the platform top, margin, slope and basin, and the positions of the wells used in the cross-section relative to these architectural elements. The wells are picked for their good quality and continuous log data, the availability of some core to calibrate the depositional facies, and their geographic distribution in a dip-section at a reasonable well spacing (ideally 10-15 km spacing). Wells too far out of the dip plane would add complexity and obscure the dip geometries. More closely spaced wells (if available) would clutter the section and would tend not to show much significant change. A strike section model would be useful to generate as a next-step (beyond the scope of this paper) once we have high confidence on the dip section. The stratigraphic cross-section is shown in figure 4.

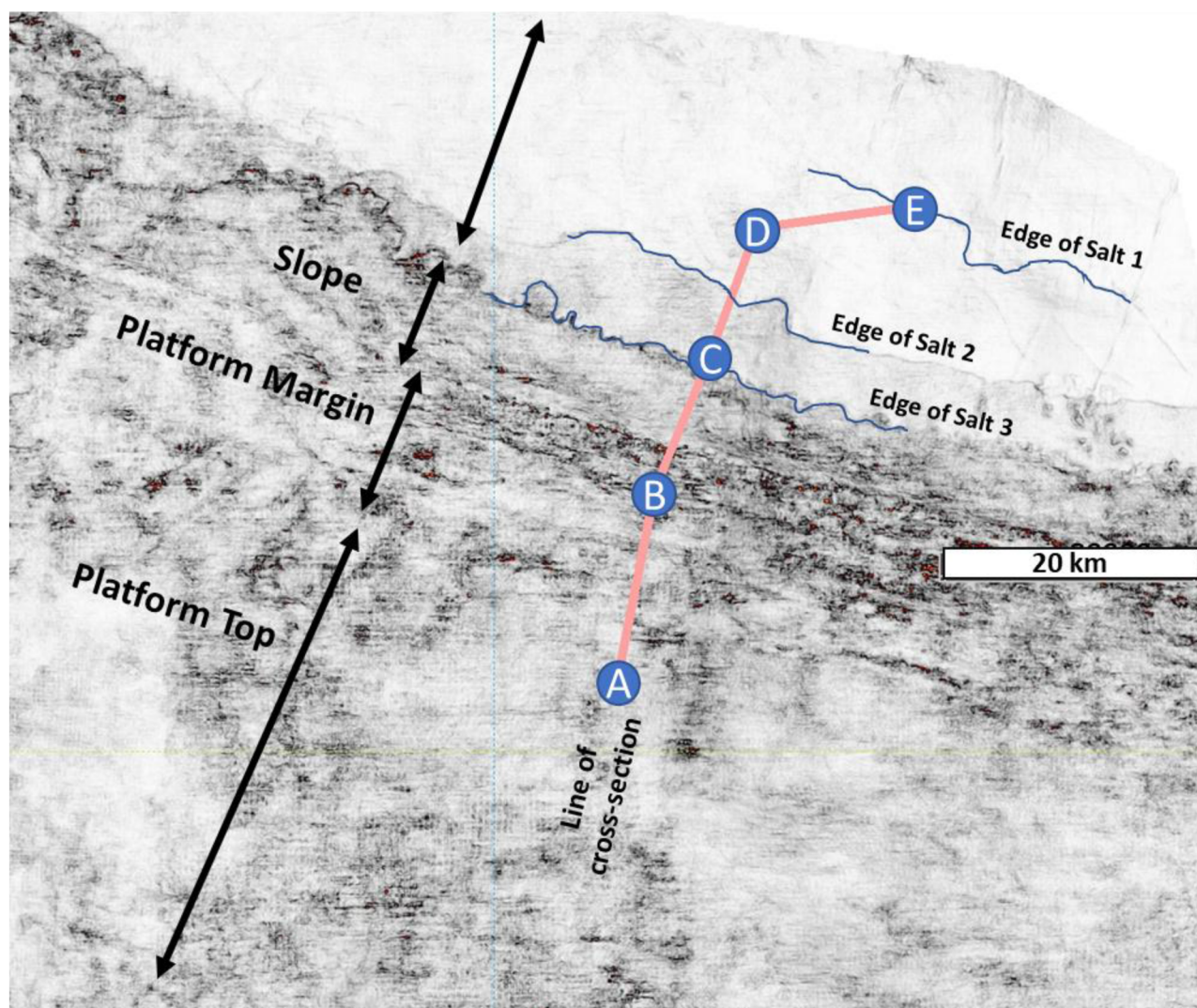


Figure 3—Detailed map showing the position of the stratigraphic cross-section relative to the platform elements. The grey-scale image is a seismic coherency slice from above the Dhruma Shales (the datum used to make the cross-section). Coherency shows the edge of salts 2 and 3 very clearly. Salt 1 is less distinct but visible (interpretation added). The platform margin and slope are evident as a set of margin-parallel lines, likely both fault related, and also capturing some of the prograding slope. The platform top is relatively featureless.

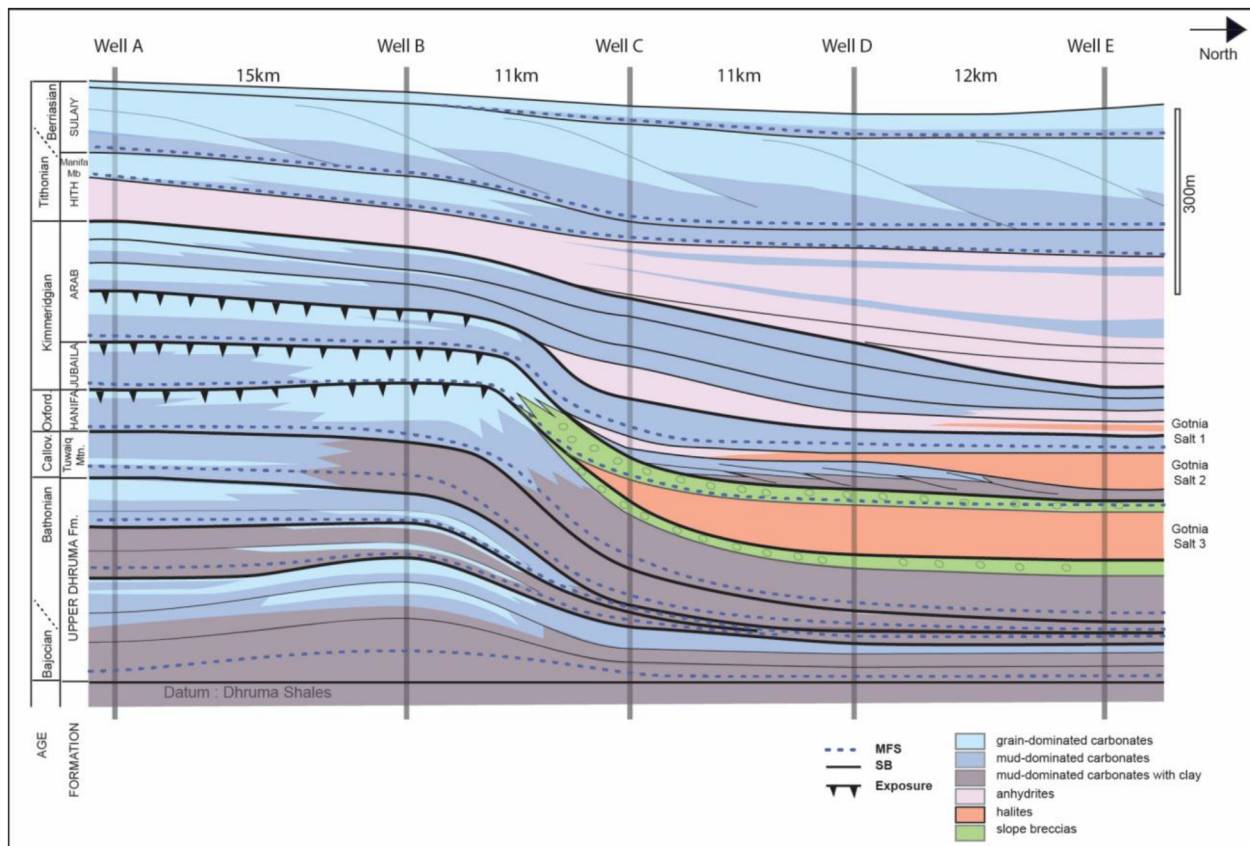


Figure 4—Stratigraphic cross-section across the Gotnia Margin. The well locations relative to the margin are shown in figure 3.

Once wells are chosen we classify the facies. Here a very simple scheme has been chosen: we distinguish grain-dominated from mud-dominated carbonates. Based on our experience with the wells in this area, we can make this interpretation from logs with reasonable confidence. Additionally, we distinguish halites (from logs), anhydrites (from logs) and slope breccias (dominantly from core, with some log characteristics). Conventional core in multiple wells across the section was reviewed to benchmark the logs and understand the depositional processes.

The next step is to define sequences at the wells: this is initially a one-dimensional interpretation which is refined as we correlate between wells. A sequence boundary on the platform from logs is typically the top of a porous unit, with an overlying flooding event into muddier carbonates. With spectral gamma we can refine this interpretation as the true sequence boundary is sometimes found at the top of a muddy terrigenous section (topsets, with a high potassium-thorium signature), which is overlain by a muddy carbonate section (high Uranium signature due to organic matter in the carbonates, but low terrigenous content). We interpret terrigenous input to represent a maximum of regression. Surfaces such as the top Hanifa Formation have been reported elsewhere to show evidence for exposure, but exposure events in the Upper Jurassic are often poorly expressed (if present) due to the arid climate. Maximum flooding events in the basinal wells are typically expressed as major gamma increases, mostly due to Uranium in organic material. These tend to appear abruptly and clean up gradually as grains are shed from the platform and dilute the organic signature.

The sequence stratigraphic significance of anhydrites is variable, since anhydrites can be sub-aqueous (often laminated, sometimes detrital or with distorted bedding) or sub-aerial (typically chicken-wire textures of nodular anhydrite growing in an early-diagenetic setting). Core is required to distinguish this (or possibly image logs), and multiple cores exist in the distal wells. Halites are readily distinguishable from logs, and can be related to significant restriction. Sylvite occurs within the halite (identified from a high gamma signature, and confirmed with XRD on cuttings samples). When correlating the sequences, we follow sequence

boundaries from platform to basin (where their expression decreases). Conversely, we trace maximum flooding surfaces from basin to platform. Seismic geometries are used to guide the interpretation and control the form of the slope.

Results

Margin architecture

1. **Sequence correlations:** salts are interpreted as lowstand deposits, evidenced by intermittent presence of sylvite within the salt, indicating complete dessication of the basin. Salt three (the lowest halite interval in the cross-section) is correlated to the Hanifa sequence boundary. The underlying sediment is Oxfordian in age (Hanifa Formation equivalent), the seismic reflectors show onlap of salt three onto a Hanifa margin. The Oxfordian is the time of maximum topographic relief at the margin. Elsewhere on the platform (beyond this cross-section), thin evaporite beds are found just beneath the top Hanifa, and this is consistent with the onset of evaporitic depositional processes in the region. Biostratigraphically a major time hiatus (the Late Oxfordian-Kimmeridgian Unconformity) is recognized on the platform, making it a very good candidate for a major sequence boundary. Heterolithic slope breccias are observed in core and are onlapped by salt three. These slope deposits are coeval with a time of falling sea-level and may have been initiated by exposure of the platform margin.
2. **Salt two (the middle halite layer)** has been correlated with the Jubaila sequence boundary on the platform. A debris apron is observed in core. The debris apron is capped by a microbial mudstone and stromatolite to sabkha assemblage, with porosity (solution-enhanced vugs, interparticle and shelter porosity) preserved in the stromatolitic units. These facies onlap the top of the debris fans in a down-dip location and pinch out down-dip. They have been interpreted to be marginal marine stromatolitic banks grading up-dip to sabkha peritidal deposits characterized by interbedded microbial mudstones with chickenwire anhydrites. The environment at this time would be hypersaline and very hostile to most organisms, with only stromatolites being able to support these salinities. These stromatolitic layers are typically thin (around 1 ft or less) but reoccur at intervals throughout the evaporitic complex in the basin.
3. **The carbonate depositional geometries** develop from a ramp morphology in the Bajocian into a high-rise platform margin in the Oxfordian, returning to a ramp geometry by the Tithonian. The development of this topography leads to the formation of the Gotnia Basin, an intrashelf basin that is then partially infilled with a thick evaporitic complex during times of marine restriction. The highstand conditions and platform morphologies that prevail during the Oxfordian are also a time of high faunal diversity, with a rich coral and reefal fauna observed in the Hanifa Formation. The older Bathonian interval is mostly dominated by coated grain deposits and mud-producing organisms, producing muddy carbonate ramps with less grain-dominated sediments.
4. **Debris aprons** with a heterolithic composition, including reworked evaporites (now anhydrite) develop during the highstand conditions of the Oxfordian, coeval with the construction of a platform margin and slope. The exact nature of the process that produces these gravity deposits is uncertain, but the debris aprons are related to times of late highstand, and are frequently onlapped by the lowstand evaporitic deposits, so it seems likely that this platform margin erosion and collapse is related to margin exposure and relative sea-level fall. Additionally, it is noted that resedimented anhydrites can be a pitfall for interpretation of depositional environments from core, because they can often resemble chickenwire textures, leading to a misinterpretation as a peritidal environment.

Basin Desiccation

The presence of sylvite in layers (from XRD measurements in cuttings taken from evaporitic intervals that show a very high gamma-ray response in logs) within the halite indicates complete desiccation of the basin at times. It is problematic to explain this desiccation with simple sea-level change, as there is over 250m of topography from platform top to basin floor, and large sea-level changes such as this have not been reported in the greenhouse world of the Late Jurassic. An alternative explanation is that smaller eustatic sea-level falls effectively isolate the basin, cutting off seaways connecting the Gotnia Basin to the open ocean, and allowing total desiccation to occur. This might be a similar process to the Mediterranean during the Messinian events.

Porosity development in the basin

Stromatolite banks are observed from core to develop in hypersaline environments during the lowstands. There is a microbiolite mud – stromatolite - evaporite facies assemblage that develops during these lowstands, interbedded with halites. These stromatolites, although thin, are important because they allow preservation of porosity in what is dominantly a tight basinal section dominated by mudstones and evaporites.

The Gotnia Margin topography

forms due to self-organizing depositional processes, not due to any significant structurally controlled subsidence. The gross isopach from Cretaceous to Mid-Jurassic shows the Gotnia Basin topography heals over the high relief platform topography that forms by the Oxfordian, leaving little evidence of variable subsidence due to faulting. Locally faults may control the location of the margin, as they create slope-breaks and irregularities that form a nucleation site for margin formation. Margin-parallel fault trends can be seen in the seismic coherency slice in [figure 3](#). But these faults do not have significant throw and do not create little topography or variations in subsidence patterns.

The platform-to-basin correlation

presented here ([figure 4](#)) is our preferred alternative, but we recognize that other correlations do exist. The platform-to-basin correlation is non-unique, due to the limitations of the basinal biostratigraphic control, and the limited resolution of the seismic data across the margin. The fundamental question is whether to correlate the Gotnia salts to the Arab cycles (Kimmeridgian) or whether the third salt correlates to the top Hanifa sequence boundary (our preferred model). Seismic correlation suggests the third salt correlates to the top Hanifa. On the platform the Hanifa shows the first onset of evaporites in the Middle to Late Jurassic. Correlating salt three with the Arab-D would create an unfeasibly high topographic relief of over 400m from platform to basin. Additionally, the Oxfordian Hanifa platform facies suggest this is a maximum for platform diversity and shelf-margin frame-builders. This is consistent with a topographic maximum.

Conclusions

Forced lowstands due to desiccation – not eustatic sea-level variations - are a mechanism for having large sea-level falls within the Gotnia Basin. During these sea-level lowstands stromatolites develop and are preserved as thin biostromal banks interbedded with evaporitic mudstones and halite layers. The development of these stromatolites provides a mechanism for the preservation of porosity in an otherwise tight and mud-evaporite dominated basinal section.

The Gotnia Basin seems to be a self-organizing intrashelf basin, developing spontaneously during a time of rapid transgression, with the basin margins formed by high-rise carbonate platform margins rather than by tectonic and subsidence processes. We know this because the gross isopach from the Early Jurassic to the Early Cretaceous shows little evidence for differential subsidence across the margin. Faulting does guide the position of the margin but it is not a primary control on its formation.

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